

ER MODEL

ENTITY-RELATIONSHIP MODEL

- **Data Model:** Collection of conceptual tools for **describing data, data relationships, data semantics, and consistency constraints.**
- **ER Model**
 - ✓ It is a high level data model based on a perception of a **real** world that consists of a collection of basic objects, called **entities** and of **relationships** among these objects.
 - ✓ Graphical representation of ER Model is **ER diagram**, which acts as a **blueprint** of DB.

ENTITY

An Entity is a “**thing**” or “**object**” in the real world that is **distinguishable** from all other objects.

- It has **physical existence**.
- Each student in a college is an entity.
- Entity can be **uniquely** identified. (By a primary attribute, aka Primary Key)
- **Strong Entity**: Can be uniquely identified.
- **Weak Entity**: Can't be uniquely identified., depends on some other strong entity.
 - ✓ It doesn't have sufficient attributes, to select a uniquely identifiable attribute.
 - ✓ Loan -> Strong Entity, Payment -> Weak, as instalments are sequential number counter can be generated separate for each loan.
 - ✓ **Weak entity depends on strong entity for existence.**

ENTITY SET

- It is a set of entities of the **same** type that share the **same** properties, or attributes.
- E.g., Student is an entity set.
- E.g., Customer of a bank

ATTRIBUTES

- An entity is represented by a set of attributes.
- Each entity has a value for each of its attributes.
- For each attribute, there is a set of permitted values, called the domain, or value set, of that attribute.
- E.g., Student Entity has following attributes
 - Registration Number
 - Name
 - Year
 - Specialization
 - Branch
 - Contact number
 - Address

TYPES OF ATTRIBUTES

➤ Simple

- Attributes which can't be divided further.
- E.g., Customer's account number in a bank, Student's Registration number etc.

➤ Composite

- Can be divided into subparts (that is, other attributes).
- E.g., Name of a person, can be divided into first-name, middle-name, last-name.
- If user wants to refer to an entire attribute or to only a component of the attribute.
- Address can also be divided, street, city, state, PIN code.

TYPES OF ATTRIBUTES

➤ Single-valued

- Only one value attribute.
- e.g., Student ID, loan-number for a loan.

➤ Multi-valued

- Attribute having more than one value.
- e.g., phone-number, nominee-name on some insurance, dependent-name etc.
- Limit constraint may be applied, upper or lower limits.

➤ Derived

- Value of this type of attribute can be derived from the value of other related attributes
- e.g., Age, loan-age, membership-period etc.

TYPES OF ATTRIBUTES

➤ NULL Value

- An attribute takes a null value when an entity does not have a value for it.
- It may indicate “not applicable”, value doesn’t exist. e.g., person having no middle-name
- It may indicate “unknown”.
 - Unknown can indicate missing entry, e.g., name value of a customer is NULL, means it is missing as name must have some value.
 - Not known, salary attribute value of an employee is null, means it is not known yet.

RELATIONSHIPS

- Association among two or more entities.
- e.g., Person has vehicle, Parent has Child, Customer borrow loan etc.
- Strong Relationship, between two independent entities.
- Weak Relationship, between weak entity and its owner/strong entity.
 - e.g., Loan <instalment-payments> Payment.

DEGREE OF RELATIONSHIP

- Number of entities participating in a relationship.
- Unary, Only one entity participates. e.g., Employee manages employee.
- Binary, two entities participates. e.g., Student takes Course.
- Ternary relationship, three entities participates. E.g, Employee works-on branch, employee works-on job.
- Binary are common.

RELATIONSHIPS CONSTRAINTS

- Mapping Cardinality / Cardinality Ratio
- Participation Constraints

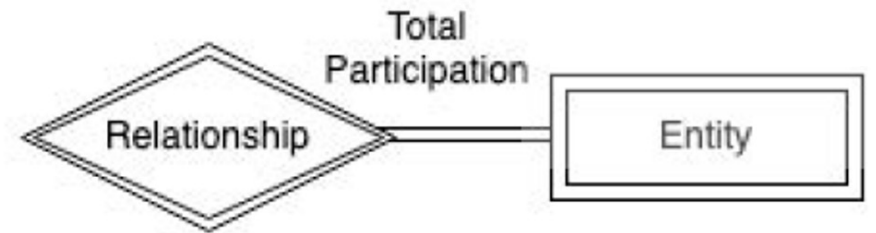
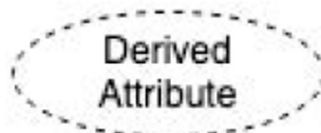
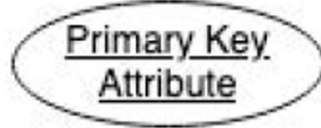
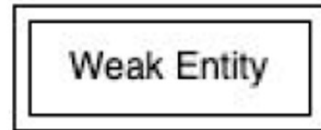
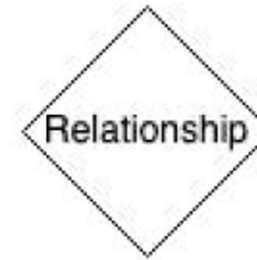
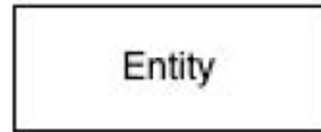
MAPPING CARDINALITY / CARDINALITY RATIO

- Number of entities to which another entity can be associated via a relationship.
- **One to one**, Entity in A associates with at most one entity in B, where A & B are entity sets. And an entity of B is associated with at most one entity of A.
 - e.g., Citizen has Aadhar Card.
- **One to many**, Entity in A associated with N entity in B. While entity in B is associated with at most one entity in A.
 - e.g., Citizen has Vehicle.
- **Many to one**, Entity in A associated with at most one entity in B. While entity in B can be associated with N entity in A.
 - e.g., Student mentored by Faculty.
- **Many to many**, Entity in A associated with N entity in B. While entity in B also associated with N entity in A.
 - Customer buys product.
 - Student attend course.
 - Course taken by professor

PARTICIPATION CONSTRAINTS

- Aka, Minimum cardinality constraint.
- Types, Partial & Total Participation.
- Partial Participation, not all entities are involved in the relationship instance.
- Total Participation, each entity must be involved in at least one relationship instance.
- e.g., Customer borrow loan, loan has total participation as it can't exist without customer entity. And customer has partial participation.
- Weak entity has total participation constraint, but strong may not have total.

SYMBOLS USED IN ER DIAGRAM



RULES FOR CREATING ER DIAGRAM

- Find the Entities and their types (Strong, Weak)
- Find all the attributes of each entity set and identify their type.(simple, composite, multivalued, derived etc.)
- Identify the relationships among the entity sets. Find their cardinality and participation.